

WMO First Mile Data Management

World Meteorological Organization

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i. Abstract

WMO regulates the real-time international exchange of observation data to ensure the availability of high-quality observational data for many applications in particular for meteorological and hydrological forecasts, and allowing Prediction Centers (for Numerical Weather Prediction – NWP as well as climate and hydrological prediction) to be able to produce reliable products. National Meteorological and Hydrological Services (NMHSs) use NWP products to monitor and predict the weather and hydrological conditions to issue forecasts and warnings for the protection of the lives and property of citizens against natural hazards, notably in the framework of the Early Warnings for All initiative.

NMHSs operate networks comprising different observing systems to meet the observational data requirements for national, regional and international needs, and to exchange data internationally, in accordance with the WMO technical regulations. Data are collected from the observing platforms, processed and quality controlled before being used nationally and exchanged internationally. WMO does not regulate this initial data collection segment of transmission of data from the observing station to the data collection and processing system (first-mile data collection).

This document defines the encoding and transmission of observation data from remote monitoring stations (Nodes) to a central data collection system (Hub).

ii. Keywords

The following are keywords to be used by search engines and document catalogues.

wmo, observations, first mile, node, hub, protobuf, mqtt

iii. Security Considerations

TODO

No security considerations have been made for this standard.

Chapter 1. Scope

This document defines the encoding and transmission of observation data from remote monitoring stations (Nodes) to a central data collection system (Hub).

This specification defines the conformance requirements for WMO First Mile Data Management (data model, encoding, transmission). Annex A defines the abstract test suite.

[1] <https://community.wmo.int/governance/commission-membership/commission-observation-infrastructures-and-information-systems-infcom/commission-infrastructure-officers/infcom-management-group/standing-committee-information-management-and-technology-sc-imt>

[2] <https://community.wmo.int/governance/commission-membership/infcom>

Chapter 2. Conformance

Conformance with this standard shall be checked using the tests specified in Annex A (normative) of this document.

WMO First Mile Data Management defines the encoding and transmission of observation data from remote monitoring stations (Nodes) to a central data collection system (Hub).

This standard identifies numerous Requirements Classes which define the functional requirements.

The mandatory Requirements Classes for this specification are:

- "Data Model"
- "Protocol"

Chapter 3. References

- Google LLC: Protocol Buffers Documentation (2026) ^[1]

[1] <https://protobuf.dev>

Chapter 4. Terms and definitions

This document uses the terms defined in [OGC Policy Directive 49](#), which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this Standard and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications ([OGC 08-131r3](#)), also known as the 'ModSpec'. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

The following additional terms and definitions also apply.

4.1. Node

The "thing" that collects and transmits data. Typically at the remote site, such as a weather station. This would be a device such as a datalogger or a sensor.

4.2. Host

The software at the central server that receives data from the Node and prepares it for further processing. It must be able to receive the data using the agreed Data Format and Protocol.

4.3. Protocol

The IP based set of rules that determine how data is transferred from the Node to the Host.

4.4. Data Format

A data format is the organisation of information according to specifications. It defines how the data is structured and encoded.

4.5. Data Serialization

The process of converting structured data into a Data Format that can be easily transmitted and reconstructed later.

4.6. Data Schema

A formal description in the declarative language defining the structure and constraints for the Data Format.

4.7. Observation

A single measurement data point collected by the Node, such as a temperature reading, humidity level, or wind speed.

4.8. Metadata

Additional information, such as the monitoring site ID, location, types of sensors, measured variables, measurement units, and other relevant details. It provides context for the data being transmitted.

4.9. Abbreviated terms

Table 1. Symbols and abbreviated terms

Abbreviation	Term
CF	Climate and Forecast (CF) Metadata Conventions
HMEI	HydroMeteorological and Environmental Industry
MQP	Message Queuing Protocol
MQTT	Message Queuing Telemetry Transport
PubSub	Publish / Subscribe
QoS	Quality of Service
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
WIS	WMO Information System
WMO	World Meteorological Organization

Chapter 5. Conventions

This section provides details and examples for any conventions used in the document. Examples of conventions are symbols, abbreviations, use of JSON Schema, or special notes regarding how to read the document.

5.1. Identifiers

The normative provisions in this Standard are denoted by the URI:

<http://wis.wmo.int/spec/fmdm/1>

All requirements and conformance tests that appear in this document are denoted by partial URIs which are relative to this base.

5.2. Examples

Protocol topic examples provided in this specification are encoded as **plain text strings**.

Data and metadata message examples provided in this specification are encoded as JSON.

Complete examples can be found at <https://schemas.wmo.int/fmdm/1.0.0/examples>

5.3. Schemas

Data and metadata message schemas can be found at <https://schemas.wmo.int/fmdm/1.0.0>

5.4. Schema representation

Protobuf is used throughout this standard to define the structure of data and metadata messages.

Data and metadata message instances are always defined as JSON.

Chapter 6. Introduction

TODO

Chapter 7. Data Model

The data model defines the structure and organization of information transmitted from the Node to the Host using Protocol Buffers (Protobuf) as the Data Format. The data model is formally specified in the protobuf schema `firstmile.proto`.

The data model supports two primary use cases:

- **Transmission of observation data:** The Node transmits measurement values collected from sensors
- **Transmission of metadata:** The Node transmits information about the monitoring site, devices, and parameter definitions

These use cases are supported by two top-level message types: the Data message containing observations, and the Metadata message containing configuration and contextual information. Both messages are defined in a single protobuf schema and transmitted as independent payloads over MQTT. The Metadata message provides the necessary context for interpreting the Data message, including descriptions of devices and parameters.

7.1. Requirements Class "Data Model"

7.1.1. Overview

This Requirements Class provides requirements for the Data Model.

The data model organizes information hierarchically as follows:

- **Messages:** Top-level containers (Data and Metadata) that are transmitted as independent payloads
- **Devices:** Descriptions of physical equipment including the Node Device (node) and Observer Devices (sensors)
- **Observations:** Measurement data points consisting of a parameter identifier, timestamp, and one or more values
- **Parameter definitions:** Metadata describing what is being measured, including units, aggregation methods, and standard name mappings. Parameter definitions also describe how observations are structured in Data messages.
- **Values:** The actual measurement data, supporting multiple numeric types, strings, booleans, and explicit representation of missing data

Each Observation references a ParameterDefinition by identifier. Each ParameterDefinition contains one or more Parameter specifications that describe the measured variable, its unit, temporal aggregation method, and which device performs the measurement.

Requirements Class
http://wis.wmo.int/spec/fmdm/1/req/data-model

Target type	Message encoding
Dependency	Protobuf
Pre-conditions	The event message conforms to the Protobuf specification.

7.1.2. Design principles

The data model is designed according to the following principles:

- **Efficiency:** Optimized for bandwidth-constrained environments but focusing on IP communication rather than extremely low-level binary formats
- **Extensibility:** Support for multiple value types and flexible parameter definitions
- **Self-description:** Metadata provides complete context for interpreting observations
- **Explicit missing values:** Clear distinction between unavailable data and zero/false measurements
- **Batch transmission:** Multiple observations may be included in a single Data message
- **Standard name interoperability:** Mapping to external naming conventions through namespace definitions

7.1.3. FirstMileMessage

A FirstMileMessage message denotes whether the content describes data or metadata.

Requirement 1	/req/data-model/first_mile_message
A	A FirstMileMessage message SHALL be used as the root definition of a message.
B	A FirstMileMessage message SHALL declare data or metadata .

7.1.4. Location

The Location message specifies the geographic position of a device (either Node or Observer).

Requirement 2	/req/data-model/location
A	A Location message SHALL define a heightMetres element in metres, above the referenceSurface .
B	A Location message SHALL define a referenceSurface element using the ReferenceSurface enumeration defined in Annex B.

Recommendation 1	/rec/data-model/location
A	A Location message SHOULD provide a latitude element describing the latitude in decimal degrees using the WGS84 coordinate system and datum. // TODO validate precision

B	A Location message SHOULD provide a longitude element describing the longitude in decimal degrees using the WGS84 coordinate system and datum. // TODO validate precision
---	--

Permission 1	/per/data-model/location
A	A Location message MAY omit the latitude and longitude elements when they are unknown.

7.1.5. Metadata

A Metadata message provides configuration and contextual information about the Node, including descriptions of devices and the parameters they measure. This message shall be transmitted at least once after Node initialization and whenever the configuration changes.

Requirement 3	/req/data-model/metadata
A	A Metadata message SHALL define a node element to describe the Node device (datalogger or station).
B	A Metadata message SHALL define one or more observers to describe individual sensor devices connected to the Node.
C	A Metadata message SHALL define one or more parameterDefinitions to define the parameters used in observations.
D	A Metadata message SHALL define a namespace element to define a mapping of namespace prefixes to their corresponding URIs for standard name resolution.

7.1.6. Metadata.Node

A Node message describes the Node device itself, typically a datalogger or observation station.

Requirement 4	/req/data-model/node
A	A Node message SHALL define a name element describing the name or model designation of the Node device.

Recommendation 2	/rec/data-model/node
A	A Node message name SHOULD contain a human-readable identification of the device, such as its model name or type designation.
B	A Node message SHOULD define a location element to describe the geographic location of the Node device.

Permission 2	/per/data-model/node
---------------------	-----------------------------

A	A Node message MAY define a url element providing a URI reference to additional metadata about the device.
B	A Node message MAY define a serialNumber element describing the manufacturer's serial number of the device.
C	A Node message MAY define a firmwareVersion element describing the firmware version of the device, typically in format such as 5.1 or 2.7.1-alpha .

7.1.7. Metadata.ObserverDevice

An ObserverDevice message describes an individual sensor or measurement device connected to the Node Device.

Requirement 5	/req/data-model/observer_device
A	An ObserverDevice message SHALL define an id element.
B	A ObserverDevice message SHALL define a name element to describe the name or model designation of the observer device.

Recommendation 3	/rec/data-model/observer_device
A	An ObserverDevice message SHOULD define a location element describing the geographic location of the observer device.

Permission 3	/per/data-model/observer_device
A	An ObserverDevice message MAY define a url element providing a URI reference to additional metadata about the device.
B	An ObserverDevice message MAY define a serialNumber element describing manufacturer's serial number of the device.
C	An ObserverDevice message MAY define a firmwareVersion element describing firmware version of the device.

IMPORTANT

Observer device identifiers shall remain stable across Node restarts and metadata updates to ensure consistent parameter-to-device associations.

7.1.8. Metadata.ParameterDefinition

A ParameterDefinition groups one or more related measurement parameters that share common context. Each parameter definition has a unique identifier that is referenced by observations. Parameters are typically grouped when they are measured at the same timestamp, allowing multiple values to share a single timestamp in observations and thus reducing payload size.

The parameters field contains one or more Parameter messages describing individual measurable variables. When multiple parameters are grouped in a single definition, each observation referencing this definition shall provide values in the same order as the parameters are defined.

Requirement 6	/req/data-model/parameter_definition
A	A ParameterDefinition message SHALL define an id element.
B	A ParameterDefinition message SHALL define a description element.
C	A ParameterDefinition message SHALL define one or more parameters a description element.

7.1.9. Metadata.ParameterDefinition.Parameter

The Parameter message specifies a single measurable variable, including its name, unit of measurement, aggregation method, and associated device.

The standardNames field enables mapping to external naming conventions such as Climate and Forecast (CF) standard names. This allows parameters to be referenced by different standard parameter types according to the target application of the device using the protocol. The Node is not bound to a specific standard but may include mappings to multiple standards as appropriate for the intended use case. The keys in this map shall correspond to namespace prefixes defined in Metadata namespaces.

The device field specifies which device (either the node or a specific observer) performs this measurement.

Requirement 7	/req/data-model/parameter
A	A Parameter message SHALL define a longName element describing the name of the parameter (e.g., "Air Temperature", "Wind Speed").
B	A Parameter message SHALL define a unit element using the WMO Register: Code Table C-6: List of units for TDCFs . // TODO verify
C	A Parameter message SHALL define a standardNames map which describes one to many namespace prefix and URI maps for the equivalent parameter name in an external vocabulary.
D	A Parameter message SHALL define a device element describing the device (Node or Observer) that measures this parameter.
E	A Parameter message SHALL define a cellMethod element describing the statistical or aggregation method applied to the measurement.
F	A Parameter message SHALL define a cellPeriodSeconds element describing the period of aggregation in seconds (0 for instantaneous point measurements).

G	When cellMethod is POINT, the value SHALL be 0, indicating an instantaneous measurement. For all other cell methods, the value SHALL be greater than 0, indicating the duration of the aggregation period. For example: * A 30-second mean temperature would have a cellMethod value of MEAN and a cellPeriodSeconds value of 30. An instantaneous wind direction would have a cellMethod value of POINT1 and a cellPeriodSeconds value of 0. A 5-minute maximum wind gust would have a cellMethod of MAXIMUM and a cellPeriodSeconds value of 300.
H	When multiple parameters are grouped in a single definition, each observation referencing this definition SHALL provide values in the same order as the parameters are defined.

7.1.10. Metadata.ParameterDefinition.Parameter.DeviceRef

The DeviceRef message associates a parameter with the device that measures it. This message uses a oneof field to reference either the node device or a specific observer device.

Requirement 8	/req/data-model/device-ref
A	A DeviceRef message SHALL one and only one of an observerId element as a reference to an ObserverDevice message or a node if the parameter is measured by the Node.
B	When the node element is present, the parameter SHALL be measured by the Node Device itself (such as internal voltage or temperature of a datalogger).
C	When the observerId field is set, it SHALL match the id of an ObserverDevice defined in the Metadata message.

7.1.11. Data

The Data message is the primary container for transmitting observation data from the Node to the Host. This message may be transmitted as frequently as needed to deliver current or batched observations.

A Data message may contain multiple Observation messages, enabling efficient batch transmission of measurement data. Each observation is independent and contains its own timestamp and parameter reference.

NOTE

The Node may transmit Data messages containing observations that reference parameter definitions not yet received by the Host. In this case, the Host shall buffer or queue such observations until the corresponding Metadata message is received.

Requirement 9	/req/data-model/data
A	A Data message SHALL define one or more observations elements to describe observation measurements.

7.1.12. Observation

The Observation message represents a single measurement data point collected at a specific time.

Requirement 10	/req/data-model/observation
A	An Observation message SHALL define a <code>parameterDefinitionId</code> element using the id of the ParameterDefinition message.
B	An Observation message SHALL define a <code>time</code> element using <code>google.protobuf.Timestamp</code> , in UTC.
C	An Observation message SHALL define one to many <code>values</code> elements using the Value message.

NOTE The `parameterDefinitionId` element shall reference a ParameterDefinition `id` from the Metadata message. The number and type of values in the values field shall correspond to the number of parameters defined in the referenced ParameterDefinition.

NOTE The time field uses the `google.protobuf.Timestamp` format, representing time as seconds and nanoseconds since the Unix epoch (`1970-01-01T00:00:00Z`).

NOTE An observation may contain multiple values when the referenced ParameterDefinition contains multiple parameters. This approach enables grouping values that are measured at the same timestamp, requiring the timestamp to be transmitted only once for multiple values. This reduces payload size, which is particularly beneficial in bandwidth-constrained environments. For example, a parameter definition for "Internal Parameters" might include both supply voltage and internal temperature, requiring two values per observation but sharing a single timestamp.

7.1.13. Value

A Value message defines the measurement of data of different types.

Requirement 11	/req/data-model/value
----------------	-----------------------

A	A Value message SHALL be defined using one of the following data types:		
	Field	Type	Description
	floatValue	float	32-bit floating point number
	doubleValue	double	64-bit floating point number
	intValue	int32	32-bit signed integer
	unsignedIntValue	uint32	32-bit unsigned integer
	int64Value	int64	64-bit signed integer
	unsignedInt64Value	uint64	64-bit unsigned integer
	stringValue	string	Text string value
	boolValue	bool	Boolean true/false value
	emptyValue	google.protobuf.Empty	Indicates missing or unavailable data
B	When a measurement value is unavailable due to sensor malfunction, communication error, or other causes, the value SHALL be represented using the emptyValue field of type <code>google.protobuf.Empty</code> .		

The choice of value type should be appropriate for the parameter being measured. Numeric measurements typically use doubleValue or floatValue, while categorical observations may use stringValue. Boolean parameters (such as alarm states) use boolValue.

NOTE	The stringValue field is intended for text data only and shall NOT be used to transmit: - binary data, including base64-encoded binary blobs - serialized objects or data structures - structured data formats such as JSON, XML, or similar encodings
NOTE	Binary data transmission and complex structured data are outside the scope of this standard.
NOTE	Missing or unavailable measurement values shall NOT be represented as: - numeric zero (0) - NaN (Not a Number) - omission from the values array - any other numeric sentinel value
NOTE	The emptyValue field shall be used to explicitly and unambiguously indicate missing data.

This approach ensures clear distinction between actual measurements of zero and missing measurements, which is critical for correct data interpretation.

When an Observation references a ParameterDefinition containing multiple parameters, the values

in the values array shall be provided in the same order as the parameters are defined in the `ParameterDefinition.parameters` field. It is not permitted to omit a value from the middle of the array, as this would cause misalignment with the parameter definitions. If a value is unavailable, the `emptyValue` field shall be used at that position to maintain correct alignment.

Chapter 8. Protocol

TODO one sentence

8.1. Requirements Class "Protocol"

8.1.1. Overview

This Requirements Class provides requirements for the Protocol.

Unresolved directive in sections/clause_8_protocol.adoc -
include::../requirements/requirements_class_protocol.adoc[]

Annex A: Conformance Class Abstract Test Suite (Normative)

A.1. Conformance Class: Data Model

label

<http://wis.wmo.int/spec/fmdm/1/req/data-model>

subject

Requirements Class "Data Model"

classification

Target Type:Event Metadata

A.1.1. Encoding

label

/conf/data-model/encoding

subject

/req/data-model/encoding

test-purpose

Validate that a data model is encoding as Protobuf.

TODO

TODO

A.2. Conformance Class: Protocol

Unresolved directive in sections/annex_a_ats.adoc -
include::../abstract_tests/ATS_class_protocol.adoc[]

Annex B: Schemas (Normative)

NOTE

Schema documents will only be published on schemas.wmo.int once the standard has been approved.

B.1. Data Model

```
syntax = "proto3";

package int.wmo.wis.spec.fmdm.1.conf.data_model;

import "google/protobuf/empty.proto";
import "google/protobuf/timestamp.proto";

enum CellMethod {
    CELL_METHOD_UNSPECIFIED = 0; // Required default
    POINT = 1; // Instantaneous value at a point
    SUM = 2; // Sum over the cell
    MAXIMUM = 3; // Maximum value
    MAXIMUM_ABSOLUTE_VALUE = 4; // Maximum of absolute values
    MEDIAN = 5; // Median value
    MID_RANGE = 6; // Average of maximum and minimum
    MINIMUM = 7; // Minimum value
    MINIMUM_ABSOLUTE_VALUE = 8; // Minimum of absolute values
    MEAN = 9; // Mean (average) value
    MEAN_ABSOLUTE_VALUE = 10; // Mean of absolute values
    MEAN_OF_UPPER_DECILE = 11; // Mean of the upper group of data values defined by
the upper tenth of their distribution
    MODE = 12; // Mode (most common value)
    RANGE = 13; // Absolute difference between maximum and minimum
    ROOT_MEAN_SQUARE = 14; // Root mean square (RMS)
    STANDARD_DEVIATION = 15; // Standard deviation within the cell
    SUM_OF_SQUARES = 16; // Sum of squares
    VARIANCE = 17; // Variance within the cell
}

message DeviceRef {
    oneof target {
        google.protobuf.Empty node = 1; // if field is present => refers to the single
node
        uint32 observerId = 2;
    }
}

message Parameter {
    string longName = 1;
    string unit = 2;
    map<string, string> standardNames = 3;
```



```

DeviceRef device = 4;
CellMethod cellMethod = 5;
uint32 cellPeriodSeconds = 6;
}

message ParameterDefinition {
  uint32 id = 1;
  string description = 2;
  repeated Parameter parameters = 3;
}

message Value {
  oneof kind {
    float floatValue = 1;
    double doubleValue = 2;
    int32 intValue = 3;
    uint32 unsignedIntValue = 4;
    int64 int64Value = 5;
    uint64 unsignedInt64Value = 6;
    string stringValue = 7;
    bool boolValue = 8;
    google.protobuf.Empty emptyValue = 9;
  }
}

message Observation {
  uint32 parameterDefinitionId = 1;
  google.protobuf.Timestamp time = 2;
  repeated Value values = 3;
}

enum ReferenceSurface {
  REFERENCE_SURFACE_UNSPECIFIED = 0; // Default
  MSL = 1;                          // Mean Sea Level
  GEOID = 2;                         // Geoid
  GL = 3;                           // Ground Level
  REFERENCE_ELLIPSOID = 4;          // Reference Ellipsoid
  PRESSURE_1000_HPA = 5;           // 1000 hPa Pressure Level
}

message Location {
  optional double latitude = 1;
  optional double longitude = 2;
  double heightMetres = 3;
  ReferenceSurface referenceSurface = 4;
}

message Node {
  string id = 1;
  string name = 1;
  optional Location location = 2;
}

```

```

optional string url = 3;
optional string serialNumber = 4;
optional string firmwareVersion = 5;
}

message ObserverDevice {
  uint32 id = 1;
  string name = 2;
  optional Location location = 3;
  optional string url = 4;
  optional string serialNumber = 5;
  optional string firmwareVersion = 6;
}

message Data {
  repeated Observation observations = 1;
}

message Metadata {
  Node node = 1;
  repeated ObserverDevice observers = 2;
  repeated ParameterDefinition parameterDefinitions = 3;
  map<string, string> namespaces = 4;
}

message FirstMileMessage {
  oneof content {
    // MUST be published with retained flag
    Metadata metadata = 1;

    // MUST NOT be published with retained flag
    Data data = 2;
  }
}

```

B.2. Enumerations

B.2.1. CellMethod

The `cellMethod` field specifies how measurement data is aggregated or processed over the cell period. The `CellMethod` enumeration defines the following values:

Name	Value	Description
<code>CELL_METHOD_UNSPECIFIED</code>	0	Required default
<code>POINT</code>	1	Instantaneous value at a point
<code>SUM</code>	2	Sum over the cell
<code>MAXIMUM</code>	3	Maximum value

Name	Value	Description
MAXIMUM_ABSOLUTE_VALUE	4	Maximum of absolute values
MEDIAN	5	Median value
MID_RANGE	6	Average of maximum and minimum
MINIMUM	7	Minimum value
MINIMUM_ABSOLUTE_VALUE	8	Minimum of absolute values
MEAN	9	Mean (average) value
MEAN_ABSOLUTE_VALUE	10	Mean of absolute values
MEAN_OF_UPPER_DECILE	11	Mean of the upper group of data values defined by the upper tenth of their distribution
MODE	12	Mode (most common value)
RANGE	13	Absolute difference between maximum and minimum
ROOT_MEAN_SQUARE	14	Root mean square (RMS)
STANDARD_DEVIATION	15	Standard deviation within the cell
SUM_OF_SQUARES	16	Sum of squares
VARIANCE	17	Variance within the cell

NOTE

When cellMethod is set to **POINT**, the measurement represents an instantaneous value and cellPeriodSeconds shall be 0. For all other cell methods, **cellPeriodSeconds** shall specify the duration over which the aggregation is performed.

B.2.2. ReferenceSurface

Name	Value	Description
REFERENCE_SURFACE_UNSPECIFIED	0	Default
MSL	1	Mean Sea Level
GEOID	2	Geoid
GL	3	Ground Level
REFERENCE_ELLIPSOID	4	Reference Ellipsoid
PRESSURE_1000_HPA	5	1000 hPa Pressure Level

NOTE

The choice of reference surface is determined by the user based on the measurement context and requirements. For example, meteorological observations of air temperature typically use height above ground level (GL), while atmospheric pressure observations may reference mean sea level (MSL).

Annex C: Examples (Informative)

C.1. Transmission topics

C.2. Data and metadata messages

Annex D: Bibliography

- IANA: Link Relation Types, <https://www.iana.org/assignments/link-relations/link-relations.xml>

Annex E: Revision History

Date	Release	Editor	Primary clauses modified	Description
2026-06-15	Template	Tom Kralidis	all	initial revision